Unsupervised Domain Adaptation for Ship Classification via Progressive Feature Alignment: From Optical to SAR Images

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***Abstract*— This article delves into the topic of unsupervised domain adaptation (UDA) by transferring knowledge from rich labeled optical domain to unlabeled synthetic aperture radar (SAR) domain, tackling the issues faced by deep-learning-based SAR ship classification methods that rely on abundant labeled SAR images. Typical UDA methods usually extract domain-invariant representations (DIRs) between two domains. However, due to the prominent differences in imaging mechanisms between optical and SAR images, the discriminative characteristics of same classes across domains may vary. Feature representation guided by labeled optical images therefore suffers from a particularly serious source-bias problem, making DIR difficult to be extracted. Moreover, capturing the category structure of the target domain is crucial for classification tasks. To solve the above challenges, this article proposes a UDA framework for SAR ship classification via progressive feature alignment between optical and unlabeled SAR domains, gradually aligning two domains across domain and class levels. At the domain level, to reduce the transfer difficulty stemming from the prominent differences between SAR and optical images, feature calibrated domain alignment (FCDA) is presented to achieve accurate DIR extraction. FCDA combines the reconstruction and the consistency constraints of different perturbed versions of the same image to calibrate the optical-bias representation into the features of unbiased toward a specific domain. At the class level, we proposed feature enhanced class alignment (FECA) to capture the fine-grained category structure of the SAR domain. FECA incorporates pseudo-label-based cross-domain contrastive learning (CDC) for intraclass compactness as well as interclass separation among cross-domain categories, along with a consistency learning approach to enhance the class structure of SAR domain. The experimental results indicate that our method achieves exceptional performance in unsupervised classification of SAR ships.**

***Index Terms*— Consistent learning, contrastive learning, feature alignment, optical and synthetic aperture radar (SAR), SAR, ship classification, source bias, unsupervised domain adaptation (UDA).**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor that can work all day and all weather. In recent years, with the development of sensors and imaging technology, SAR can quickly acquire large-scale and high-quality SAR images, making SAR occupy an increasingly important position in remote sensing applications. SAR ship classification is one of the most practical applications among the many applications of SAR. It plays a basic and key role in tasks such as ocean monitoring, maritime traffic planning, and military intelligence acquisition [1].

In the past decade, SAR ship classification methods based on deep learning have garnered significant interest and achieved superior performance [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], including the methods based on network architecture design [2], [3], [4], the methods based on feature fusion [5], [6], [7], [8], [9], and the methods based on attention [10]. Nevertheless, these approaches generally acquire a significant amount of SAR images with class annotation to train the deep classifier, which limits the application of SAR ship classification in practice [12]. Unlike optical remote sensing images, the imaging mechanism of SAR leads to a significant difference between SAR images and human visual perception. Therefore, collecting rich labeled SAR training data with fine-grained categories is often expensive and time-consuming, which has become one of the most important bottlenecks hindering to train the deep SAR ship classifier in real application scenarios [13].

Unsupervised domain adaptation (UDA) is a promising and feasible solution to solve the above problems. Unlike traditional supervised learning, which directly trains the classifier with labeled images, UDA aims to improve the unsupervised classification performance of unlabeled target domain by leveraging knowledge from labeled source domain that is different but related to target domain [14]. Currently, several UDA methods have been applied in the field of SAR target classification. Typical UDA methods usually achieve domain distribution alignment by extracting domain-invariant representations (DIRs) between two domains so that the classifier learned by labeled source domain will perform well in the target domain [15], [16], [17], [18], [19], [20]. Huang et al. [21] applied maximum mean discrepancy (MMD)

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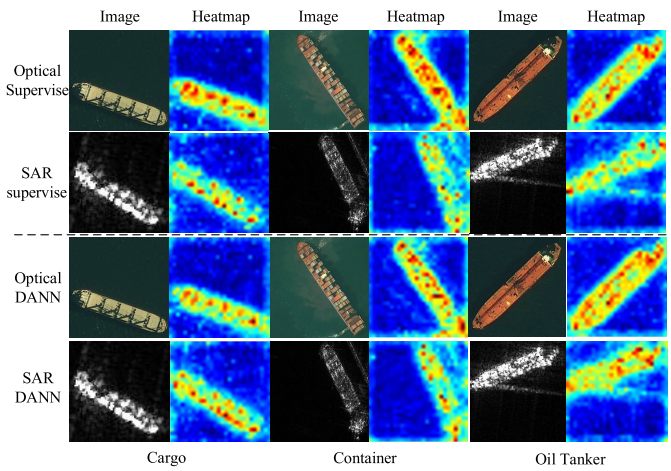


Fig. 1. Typical examples of optical and SAR remote sensing ship images and heatmaps. Each row gives the original images and the corresponding heatmaps learned by different models, which are the supervised optical model, the supervised SAR model, and the DANN model.

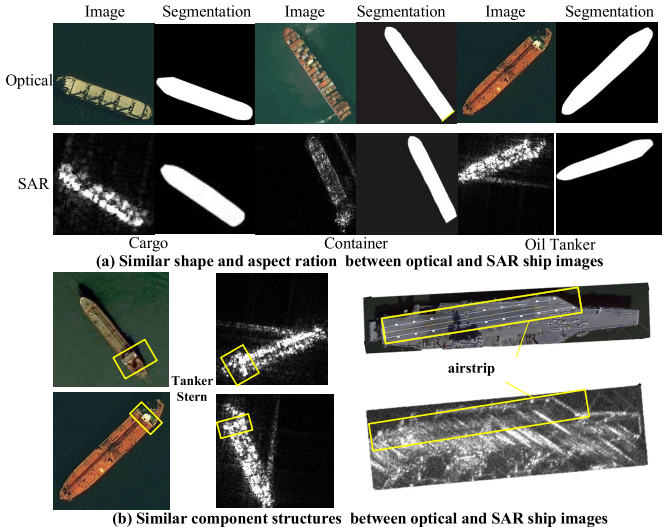


Fig. 2. Some ship similarities between the same class across optical and SAR images. (a) Illustrations of similarities in the shapes. (b) Illustrations of similarities in the local component structure.

[22] to extract DIR between multiple source domains and SAR target domain. Geng et al. [23] introduced a novel approach known as the joint MMD technique, which aims to map representations from multiple SAR spaces into common subspace to learn DIR. To extract the DIR between the simulated SAR and real SAR data, Lyu et al. [24] used both MMD and domain adversarial neural network (DANN) [25] (adversarial learning) to achieve better global alignment for unsupervised SAR image classification. Chen et al. [26] initiated the process by translating simulated SAR data to real SAR data with cycle generative adversarial network (CycleGAN) [27] and subsequently extracted DIR with adversarial learning.

The aforementioned approaches give a practical idea to improve the performance of unlabeled target domain by leveraging the knowledge of the source domain whose labeled samples are convenient and low cost to acquire. In this article, we aim to explore unsupervised SAR ship classification by transferring knowledge from the labeled optical domain to unlabeled SAR domain. Leveraging a large number of labeled optical images to assist learning of the SAR classifier is a very attractive but nascent field. Despite the pioneering works of Rostami et al. [28] and Lang et al. [29] to explore the transfer learning between the optical domain and the SAR domain for ship classification task, they require labeled SAR images to guide the classifier. To the best of our knowledge, the UDA classification research from optical to SAR images is still in the void. Compared to the supervised domain adaptation (DA) problem, the UDA problem is more challenging due to the huge differences between optical and SAR images. The main problems can be described as follows.

1) Typical UDA methods aim to extract DIR between two domains to achieve domain-level alignment by adversarial learning (such as DANN) or MMD. However, due to the prominent differences in imaging mechanisms between optical and SAR images, the discriminative characteristics of same classes across domains may vary. As shown in Fig. 1, for three ship classes in optical data, the most discriminative feature is actually the items on the deck (heatmaps reflect the focus degree of the network to different regions). Cargo ships typically transport blocky warehouses and container ships carry densely packed containers, while tankers store fuel oil in bottom bins, resulting in fewer items on the hull surface. Such discriminative features are often difficult to be displayed in SAR data with different image characteristics. Feature representation guided by labeled optical images therefore suffers from a particularly serious source-bias problem. For better extracting DIR, the model is expected to focus less on such optical-bias features and more on the information that coexisted both in optical and SAR images. Fig. 2 shows some explicit illustrations of domain-shared characteristics that are obvious and intuitive to be understood. For example, ships of the same class have similar shapes, while ships of different classes have different shapes, and some component structures of the same class exhibit the same characteristics in optical and SAR images. We are confident that there are still other implicit domain-shared characteristics across domains, which are not intuitively understandable by humans but can be automatically captured by deep models in a data-driven manner. The implicitly shared features of ships refer to the inherent physical attributions of ship targets, such as the number of hulls, whether it carries weapons, the location of the ship (e.g., a container ship may be close to a loading dock), the tonnage of the ship and its draft, and so on. However, existing UDA methods (such as DANN in Fig. 1) ignore the hindrance of optical-bias problem when forcing the features of the source and target domains to be similar, making DIR between the optical and SAR domain difficult to extract.
2) Even if the DIR can be extracted, another major limitation of previous methods is that they neglect the finer class-specific information of target domain during the alignment, i.e., intraclass compact and

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interclass separation, while learning domain-agnostic feature representations. On the one hand, samples of different classes between two domains may be aligned incorrectly. The underlying reason is that DIR constraints (e.g., MMD and DANN loss) can be optimized even when the samples of target domain are aligned with source samples of different classes. On the other hand, since the target-bias features of SAR images, including the discriminative features for SAR target classification, are not considered, the learned decision boundary may generalize poorly for the target domain.

To solve the above challenges, we proposed a novel UDA method for SAR ship classification by progressively aligning features at domain and class levels by transferring the knowledge from labeled optical images to unlabeled SAR images. At the domain level, to reduce the transfer difficulty caused by the huge differences between the SAR images and optical images, feature calibrated domain alignment (FCDA) is proposed to achieve accurate DIR extraction. On the infrastructure of adversarial-learning-based DIR, FCDA additionally combines the reconstruction and the consistency constraints of different perturbed versions of the same image to learn the features of unbiased toward specific domain, implicitly reducing the domain characteristics (i.e., optical-bias characteristics) in the representations. At the class level, we proposed feature enhanced class alignment (FECA) to learn the finer class information of the SAR domain. FECA consists of pseudo-label-based cross-domain contrastive learning (CDC) and consistency learning-based approach. The proposed CDC models the pairwise similarities between two domains to achieve intraclass compact as well as interclass separation among cross-domain categories, where the class information of SAR images is obtained with pseudo-labels. To ensure the correctness of pseudo-labels, the noise labels filtering criterion based on the size prior information of ships is designed. Intuitive example is that the overlarge ships must not be fishing ships physically. We called the "fishing ships" as complementary labels, i.e., "what it is not." We further use the consistency regularization of pseudo-label and complementary labels to enhance the class structure of unlabeled SAR images. In the experiments, this study collects a new dataset, which contains five-class ships with a total of 2505 images. The experimental results indicate that our method achieves exceptional performance in unsupervised classification of SAR ships. More specifically, the main contributions can be described as follows.

1) We proposed a novel UDA method for ship classification based on progressively aligning features at domain and class levels by transferring the knowledge from the labeled optical images to unlabeled SAR images. To the best of our knowledge, it is the first work for the UDA classification from optical to SAR domain. Besides that, for research purposes, we collect a new dataset, which contains five-class ships with a total of 2505 images.
2) To reduce the transfer difficulty caused by the prominent differences between the SAR images and optical images, FCDA is proposed to achieve accurate DIR extraction.
3) FECA is designed to learn the fine-grained category information of the SAR domain. FECA incorporates pseudo-label-based CDC for intraclass compactness as well as interclass separation among cross-domain categories, along with a consistency learning approach to enhance the class structure of SAR domain.

The remainder sections are organized as follows. Section II will discuss the related work. Section III will introduce the proposed UDA method for ship classification. Section IV will present the experimental results and analysis. Section V will give the conclusion.

II. Related Work

In this article, we will discuss related works about the DA for SAR classification and optical-to-SAR transfer in the remote sensing field.

A. DA for SAR Classification

SAR target classification networks based on deep learning often require a large amount of labeled data to effectively extract hierarchical features. However, labeling SAR images is difficult and costly, and the number of labels is difficult to achieve the same level of labeled samples as optical remote sensing images, which limits the performance of deep SAR target recognition models. Domain adaptation, as a classical transfer learning method, offers practical solutions to address the issue by transferring knowledge from a source domain with rich labels to a target domain with few/no labeled samples. Wang et al. [30] introduced a domain adaptation SAR classification approach by combining domain adversarial training and meta-learning. Based on the meta-learning training strategy, rich simulated SAR data are used to pretrain the model, and the distribution difference between simulated and real SAR images is reduced through domain adversarial learning. Bi et al. [31] introduced a DA approach to augment limited training data at the pixel level, which reduced the distribution difference through generating real SAR images from simulated SAR images. Huang et al. [32] showed that by reconstructing a significant number of unlabeled spaceborne SAR images with an autoencoder, the features can be enhanced and benefit from the SAR classification task. Huang et al. [21] applied MMD to reduce the distribution gap between multiple source domains (including nature images and SAR images) and SAR target domain, achieving better SAR classification performance. Geng et al. [23] introduced a novel approach known as the joint MMD technique, which aims to map representations from multiple SAR spaces into common subspace to learn DIR. To extract the DIR between the simulated SAR and real SAR data, Xiaoling et al. [24] used both MMD and DANN [25] (adversarial learning) to achieve better global alignment for unsupervised SAR image classification. Chen et al. [26] initiated the process by translating simulated SAR data to real SAR data with CycleGAN and subsequently extracted DIR with adversarial learning. Though the above methods extract the DIR to achieve global domain alignment, they neglect the fine-grained category-specific structure of target SAR domain in the UDA process. To solve this issue, Shi et al. [33] first attempted

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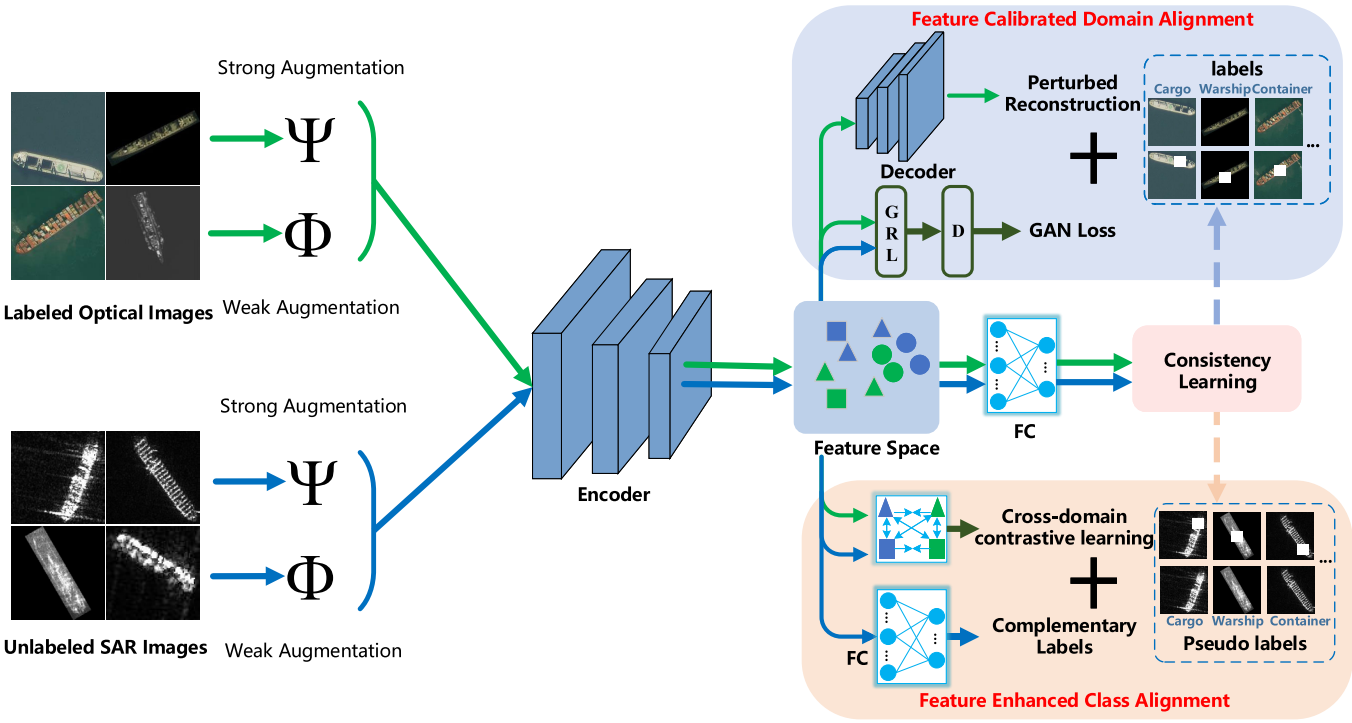


Fig. 3. Illustration of the proposed UDA framework for SAR ship classification, which gradually reduces the features gap at the domain and class level between the rich labeled optical images and unlabeled SAR images. The proposed framework includes two stages: 1) based on the infrastructure of adversarial-learning-based DIR, FCDA combines the reconstruction and the consistency constraints of different perturbed versions of the same image to learn the features of unbiased toward specific domain and 2) FECA consists of pseudo-label-based CDC and consistency learning based approach.

category-aligned UDA SAR classification, where they fused electromagnetic scattering features and deep model predictions to obtain more accurate category pseudo-labels for feature alignment. Most of the current UDA methods explore the problems of knowledge transfer between homogeneous data (SAR-to-SAR or simulation-to-reality). However, the SAR-to-SAR approaches need labeled SAR samples to train the classifier, while the simulation-to-SAR methods face the problems of electromagnetic simulation and model inaccuracy. In this article, we aim to delve into the topic of UDA for ship classification by transferring from the optical domain. Due to huge differences of imaging mechanisms between two domains, optical-to-SAR UDA faces new troubles and challenges.

B. Optical-to-SAR Transfer for Remote Sensing Data

Transferring the knowledge of richly labeled optical data to SAR data is a newly emerging field and has great potential. Compared with classification tasks, research on target detection based on optical-to-SAR transfer is more in-depth and extensive. To boost the performance of SAR target detection, Guo et al. [34] utilized large-scale optical vehicle data to assist the learning of small-scale training SAR data for detection and aligned the feature distribution of the two data domains through instance-level adversarial learning. Liao et al. [35] introduced a semi-supervised domain-adapted SAR target detection method with the help of optical remote sensing data. This method includes the MMD constraint to reduce the distribution gap among optical and SAR domains, and a reconstruction module to realize the full potential of unlabeled data. Shi et al. [13] first attempted to mine the UDA detection between optical and SAR data and improved the performance significantly. Due to the huge difference between two domains, Shi et al. gradually transferred to overcome the obstacles, including pixel stage, feature stage, and self-training stage. In this study, Shi et al. first answered the questions of what knowledge is transferred between two domains and why transferred knowledge is effective for detection task. Motivated by Shi et al., more and more UDA detection methods about optical-to-SAR transfer are occurred. Following the progressive transfer framework of Shi et al., Yuan et al. [36] utilized generative adversarial network (GAN) [37] to reduce the pixel-level differences, aligned the local and global distributions at feature level, and learned the pseudo-labels with regularization learning at prediction level. He et al. [38] progressively transferred knowledge by feature alignment at scene level, local level, global level, and instance level. Yu et al. [39] collected a new optical-to-SAR detection dataset and applied the CycleGAN to overcome the differences between optical and SAR domains with the idea of image translation. Yang et al. [40] combined the attention mechanism and CycleGAN to generate fake SAR images from optical images and designed a new feature extractor with attention module to reduce the effect of complex backgrounds.

Research on target classification based on optical-to-SAR transfer is relatively limited. Rostami et al. [28] proposed to minimize the distance of feature distribution between optical and SAR data and then use fewer labeled SAR samples to fine-tune the network for ship classification (ships or not ships).

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Lang et al. [29] introduced a multisource domain adaptation method that transfers the knowledge from the automatic identification system (AIS) domain and optical domain to SAR domain with limited labeled samples, achieving great performance. To overcome the difficulty of sparse labels and imbalanced classes in supervised SAR ship classification, Gao et al. [41] proposed an image translation network by translating the optical to fake SAR images to augment SAR data. Though optical-to-SAR transfer for classification based on supervised domain adaptation has made some progress, to the best of our knowledge, the UDA classification from optical to SAR images is still in the void. In this article, we delve into the differences between optical and SAR data for classification tasks and propose a new UDA approach for ship classification based on progressively aligning features at domain and class levels by transferring the knowledge from the labeled optical images to unlabeled SAR images. Besides that, we collect a new dataset, which contains five-class ships with a total of 2470 images for research purposes. We hope that our work can serve as a baseline and inspire further research.

III. Proposed Method

Sections III-A–III-C outlines our UDA approach for SAR ship classification, focusing on progressively aligning features. In the context of UDA SAR ship classification, the method involves utilizing a set of rich labeled optical images as the source domain $D_o = \{(x_i^o, y_i^o)\}_{i=1}^M$ and a set of unlabeled SAR images as the target domain $D_{\text{SAR}} = \{(x_j^{\text{SAR}})\}_{j=1}^N$. Adhering to inductive learning principles, the unlabeled SAR data are partitioned into training and test sets. The objective is to train a cross-domain ship classifier using labeled optical images and unlabeled SAR images that can effectively classify SAR ships.

The framework of the proposed UDA SAR ship classification is depicted in Fig. 3. The proposed framework includes two stages.

1) *First Stage:* Performing the domain-alignment to reduce the global distribution differences. Based on the infrastructure of adversarial-learning-based DIR, FCDA combines the reconstruction and the consistency constraints of different perturbed versions of the same image to calibrate the optical-bias representation into the features of unbiased toward specific domain.
2) *Second Stage:* Performing the class alignment to capture the finer class structure of SAR domain. FECA includes pseudo-label-based CDC and consistency learning-based approach. The proposed CDC models the pairwise similarities between two domains to capture fine-grained category alignment, where the class information of SAR images is obtained with pseudo-labels. Complementary labels are introduced to fully exploit unlabeled data. We further use the consistency regularization of pseudo-labels and complementary labels to enhance the class structure of unlabeled SAR images.

Next, the details of the proposed method will be introduced.

A. Feature Calibration Domain Alignment

Typical UDA methods usually extract DIR between two domains to achieve domain-level alignment. However, due to the prominent differences in imaging mechanisms between optical and SAR images, the discriminative characteristics of same classes across domains may vary. Feature representation guided by labeled optical images therefore suffers from a particularly serious source-bias problem, making DIR difficult to be extracted. FCDA aims to achieve accurate extraction of DIR between optical and SAR domains with large imaging differences by implicitly reducing the optical-bias characteristics. On the infrastructure of adversarial-learning-based DIR, FCDA additionally combines the reconstruction and the consistency constraints of different perturbed versions of the same image to learn the features of unbiased toward specific domain. Next, we will introduce the reconstruction and the consistency constraints with different perturbed versions and finally introduce the classic adversarial-based DIR extraction.

1) Perturbed Reconstruction and Consistency Constraints: As illustrated in Fig. 3, when given an optical input x, weak augmentation function Φ and strong augmentation function Ψ are applied to perform the different disturbances

$$O_i^w = \Phi(x) \tag{1}$$

$$O_i^s = \Psi(x). \tag{2}$$

The weak augmentation involves common techniques such as image flipping, while inspired by the self-supervision idea of MAE [42], we use Random_Cutout as our strong data augmentation. We feed O_i^w and O_i^s to the encoder to obtain the hidden representations, and then, the hidden representations flow into the decoder to reconstruct O_i^w and x. The above process can be optimized as follows:

$$\mathscr{L}_{\text{PR}} = \frac{1}{B}\sum_{i=1}^{B}\left(\|\text{De}(\text{En}(\Phi(x_i))) - \Phi(x_i)\|_2 + \|\text{De}(\text{En}(\Psi(x_i))) - x_i\|_2\right) \tag{3}$$

where B is the batch size of each mini-batch, En represents the encoder function, and De represents the decoder function.

The weak augmentation reconstructs the signal given its complete observation information, while the strong augmentation version reconstructs the original signal given its partial observation. The objective computes the L2 loss between the reconstructed and original images in the pixel space. The reconstructions of different perturbed versions encourage the model to focus not only on more discriminative features but also on the information contained in SAR images, such as shape, structure, and edge of different classes.

Consistency of the different perturbed versions aims to encourage the model to be robust to changes in image space so that the model is not limited to focusing only on the regions with more discriminative features. The core idea is to require the same predictions of different perturbed versions of the same image. We first input weak augmented samples to classifiers and enforce consistency against their corresponding strong augmented variant. The above process can be optimized as follows:

$$\mathscr{L}_{\text{PC}} = \frac{1}{B}\sum_{i=1}^{B}\left(H\left(p_{i,o}^w, y\right) + H\left(p_{i,o}^w, p_{i,o}^s\right)\right) \tag{4}$$

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where $H(\cdot,\cdot)$ represents the cross-entropy loss between two probability distributions and $p_{i,o}^{w}$ and $p_{i,o}^{s}$ represent the classifier's outputs of weak augmentation and strong augmentation, respectively.

2) Adversarial-Learning-Based DIR Extraction: The reconstruction and the consistency constraints of different perturbed versions of the same image implicitly reduce the optical-bias features in the representations, making DIR easier to extract between optical and SAR images. In our method, adversarial learning has been applied to extract DIR. The discriminator D, functioning as a binary classifier, is trained to determine if a feature belongs to the optical or SAR domain. The encoder (served as generator in adversarial learning) is trained to deceive the discriminator by generating ambiguous features. The encoder and discriminator are trained with an adversarial learning manner. The above process can be optimized as follows:

$$\begin{aligned}\mathscr{L}_{\mathrm{DIR}}(\mathrm{En}, D, X_{\mathrm{optical}}, X_{\mathrm{SAR}}) &= E_{x_{\mathrm{optical}}\sim X_{\mathrm{optical}}}[\log D(\mathrm{En}(x_{\mathrm{optical}}))] \\ &\quad + E_{x_{\mathrm{SAR}}\sim X_{\mathrm{SAR}}}[\log(1 - D(\mathrm{En}(x_{\mathrm{SAR}})))]. \end{aligned} \tag{5}$$

The objective of discriminator is to minimize the loss while the objective of generator is to maximize the loss. Instead of alternating iterative training, we employ the gradient reversal layer (GRL) [25] for the end-to-end adversarial training by inserting it between the encoder and discriminator. In the forward process, GRL is served as an identity transform, while in backpropagation, it reverses the gradient by multiplying a $-\alpha, \alpha > 0$ between discriminator and encoder

$$\hat{g} = -\alpha \cdot g. \tag{6}$$

The overall training objective of FCDA stage is the combination of perturbed reconstruction loss, perturbed consistency loss, and adversarial-based DIR loss

$$\mathscr{L}_{\mathrm{FCDA}} = \lambda_{\mathrm{PR}}\mathscr{L}_{\mathrm{PR}} + \lambda_{\mathrm{PC}}\mathscr{L}_{\mathrm{PC}} + \lambda_{\mathrm{DIR}}\mathscr{L}_{\mathrm{DIR}} \tag{7}$$

where λ_{PR}, λ_{PC}, and λ_{DIR} are the balance parameters. For simplicity, they are all set as 1.

3) How the Proposed Methods Reduce Source-Bias Features?: The purpose of the proposed method is to expect that the model will focus less on the more separable features of optical images but focus more on all features, making DIR easier to learn. Specifically, the proposed method includes two parts. One part is perturbation reconstruction loss and consistency loss, which reduces the network's attention to optically separable features and increases its attention to global features. The other part is the DIR based on adversarial learning. Based on the previous part, domain-invariant feature representation is extracted more easily. The DIR based on adversarial learning is a generalization of $\mathscr{H}$-divergence [43]. It introduces a domain classifier to measure the difference between two distributions, thereby reducing source-biased features and extracting domain-invariant features.

B. Feature Enhanced Class Alignment

Even if the DIR can be extracted, the fine-grained category information of target domain is neglected during the alignment while learning domain-agnostic feature representations. In this section, FECA is proposed to capture the category structure of the unlabeled SAR data. FECA incorporates pseudo-label-based CDC along with a consistency learning-based approach to enhance the class structure of SAR domain. In the following, the method of the proposed CDC and consistency learning approaches is given.

1) Pseudo-Label-Based Cross-Domain Contrastive Learning: For class alignment, we aim to minimize the discrepancy between samples with the same classes across domains while maximizing the discrepancy between the samples with the different classes. Contrastive learning is used to compact the intraclass representations and push the interclass samples away from each other

$$\mathscr{L}_{\mathrm{CDC}} = -\log \frac{\sum_{F(x_j)\in B^{i+}} e^{d(\mathrm{En}(x_i),\mathrm{En}(x_j))}}{\sum_{F(x_j)\in B^{i+}} e^{d(\mathrm{En}(x_i),\mathrm{En}(x_j))} + \sum_{F(x_j)\in B^{i-}} e^{d(\mathrm{En}(x_i),\mathrm{En}(x_j))}} \tag{8}$$

where $d(\cdot,\cdot)$ represents the Euclidean distance, B^{i+} is the positive set of x_i in optical and SAR domains, and B^{i-} is the negative set of x_i in optical and SAR domains.

However, in the UDA settings, we cannot access the class information of SAR domain directly, so we use the maximum posterior probability of classifier to assign pseudo-labels for SAR images in a mini-batch. Pseudo-labels of the first mini-batch are obtained by the FCDA classifier. Then, the pseudo-labels and model parameters in the following are then alternately updated [44]. To improve the correctness of the class alignment, we only selected the pseudo-labels for the SAR images whose maximum posterior probability exceeds a threshold and filters the noise labels combined with prior size information of ships.

The integration of prior information into machine learning has a rich history [45]. Leveraging prior knowledge helps to minimize the reliance on training data and can rectify the error induction of the model. Prior knowledge of targets can be embedded in many forms, such as statistical descriptions (boundary value and mean value of target size, shape, aspect ratio, and so on) from other data or expertise knowledge, physical principles, mathematical rules, and so on. Such prior knowledge may offer valuable clues in domain adaptation tasks, as this information does not change with domain changing. In this article, we mainly focus on the prior information of ship's size since it can be easily obtained from labeled optical data or human experts [29]. Intuitive example is that the overlarge ships must not be fishing ships physically. For simplicity, we only filter those ships that significantly deviate the physical size prior. We establish an upper bound ξ_u on the size of small ships (e.g., fishing ship) and a lower bound ξ_l on the size of large ships (e.g., container ship) through statistical descriptions derived from optical data and expert knowledge of AIS. For example, a fishing ship with a size exceeding ξ_u will be filtered. For the ships with large variations of size (e.g., cargo ship and oil tanker), we do not set any prior filtering solution. Pseudo-label filtering occurs in each mini-batch by setting a mask for samples that do not satisfy the

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Algorithm 1 Proposed Pseudo-Label Filtering Strategy Combined Confidence and Size Prior

Input: Batch of target SAR ship images with image size; Weak-augment function Φ; Encoder En; Classifier CLS; Confidence threshold T; size prior set of each class ships with upper bound or lower bound $PS\{class1(size), class2(size), \ldots\}$;

Output: Masked matrix M;

Initialize M;
for iteration $iter$ do:
  for $i = 1$ to B in a target mini-batch do:
    $p_i^w = CLS(En(\Phi(x_i^t)))$
    $s_i = size(x_i^t)$
    if $\arg\max(p_i^w) > T$
      && $s_i \in PS(index(\arg\max(p_i^w)))$:
      $M[i] = 1$
    else:
      $M[i] = 0$
return M.

confidence score and size priors. The algorithm is described in Algorithm 1.

In this way, we get a masked pseudo-labels of the target domain in each mini-batch. To better perform the CDC with a domain-balance manner, we choose the top k samples of the source domain that are far distant from the target samples by calculating the Euclidean distance $d(f_i, f_j)$ between each cross-domain pair, where k equals the number of pseudo-samples in target mini-batch

$$d(f_i, f_j) = (f_i - f_j)^2. \tag{9}$$

2) Consistency Enhancement Learning: The proposed CDC approach regards the high-confident samples that do not diverge from the size prior as the pseudo-labels to train the model while preventing the exploitation of the low-confident samples or samples that diverge from the size prior. This neglect of potentially valuable unlabeled samples may compromise the final outcomes. An intuitive example is that the overlarge ships must not be fishing ships physically and the "fishing ships" will be wasted in the current mini-batch. To fully exploit these unlabeled data, we treat the "fishing ships" as complementary labels, i.e., "what it is not." Similarly, in certain scenarios, a cross-domain classifier (called positive classifier) may struggle to accurately predict "what it is." However, the classifier can more easily guess "what it is not," for example, "it is not a fishing ship" or "it is not a warship." Therefore, we also set the category index with the lowest confidence as the complementary labels for the low-confident samples. Specifically, another negative classifier is trained to exploit the complementary labels. Motivated by the popular consistent learning in semi-supervised learning literature recently, we further use the consistency regularization [46], [47], [48] of pseudo-labels and complementary labels to enhance the class structure of unlabeled SAR images. Fig. 4 shows the illustration of proposed consistency enhancement learning. For unlabeled SAR images in each mini-batch, we divide them into two parts according to their confidence and size prior: samples with pseudo-labels and samples with complementary labels.

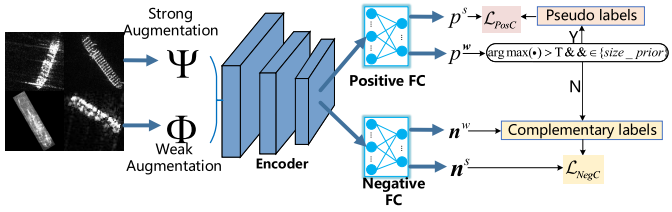


Fig. 4. Illustration of the proposed consistency enhancement learning. For unlabeled SAR images, we divide them into two parts according to their confidence and size prior: samples with pseudo-labels and samples with complementary labels. For pseudo-labels, we enforce the output of the positive classifier from different perturbed versions to be consistent. For complementary labels, we enforce consistency regularization on negative classifier with soft labels of different perturbed versions.

For pseudo-labels, we enforce the positive classifier's output of different perturbed versions to be consistent

$$\mathscr{L}_{\text{PosC}} = \frac{1}{B}\sum_{i=1}^{B} 1(x_i \in S_{\text{pseu}})\left(H\left(\hat{p}_i^w, p_i^w\right) + H\left(\hat{p}_i^w, p_i^s\right)\right) \tag{10}$$

where $1(\cdot)$ represents the indicator function; S_{pseu} is the sample set that satisfies the confidence threshold and prior assumption; $\hat{p}_i^w$ represents the pseudo-labels assigned by the positive classifier with one-hot form; p_i^w and p_i^s represent the positive classifier's outputs of weak augmentation and strong augmentation, respectively; and $H(\cdot,\cdot)$ represents the cross-entropy loss between two probability distributions. The first term is used to train the positive classifier to classify "what it is."

For complementary labels, we enforce the negative classifier's output of different perturbed versions to be consistent

$$\mathscr{L}_{\text{NegC}} = \frac{1}{B}\sum_{i=1}^{B} 1(x_i \notin S_{\text{pseu}})\left(\bar{H}\left(\hat{c}_i^w, n_i^w\right) + H\left(n_i^w, n_i^s\right)\right) \tag{11}$$

where $\hat{c}_i^w$ represents the complementary labels assigned by our method; n_i^w and n_i^s represent the negative classifier's outputs of weak augmentation and strong augmentation, respectively; and $H(\cdot,\cdot)$ represents the cross-entropy loss between two probability distributions. The first term is used to train the negative classifier to classify "what it is not." In the optimization of first term, to prevent the encoder from being affected by the training of the negative classifier, we apply the gradient isolation strategy [49] that only updates the negative classifier without updating the parameters of encoder. $\bar{H}(\cdot,\cdot)$ represents inserting the gradient isolation strategy into the optimization process.

The overall training objective of FECA stage is the combination of cross-domain contrastive loss, positive consistency enhancement loss, and negative consistency enhancement loss

$$\mathscr{L}_{\text{FECA}} = \lambda_{\text{CDC}}\mathscr{L}_{\text{CDC}} + \lambda_{\text{PosC}}\mathscr{L}_{\text{NegC}} + \lambda_{\text{NegC}}\mathscr{L}_{\text{NegC}} \tag{12}$$

where λ_{CDC}, λ_{PosC}, and λ_{NegC} are the balance parameters. For simplicity, they are all set as 1.

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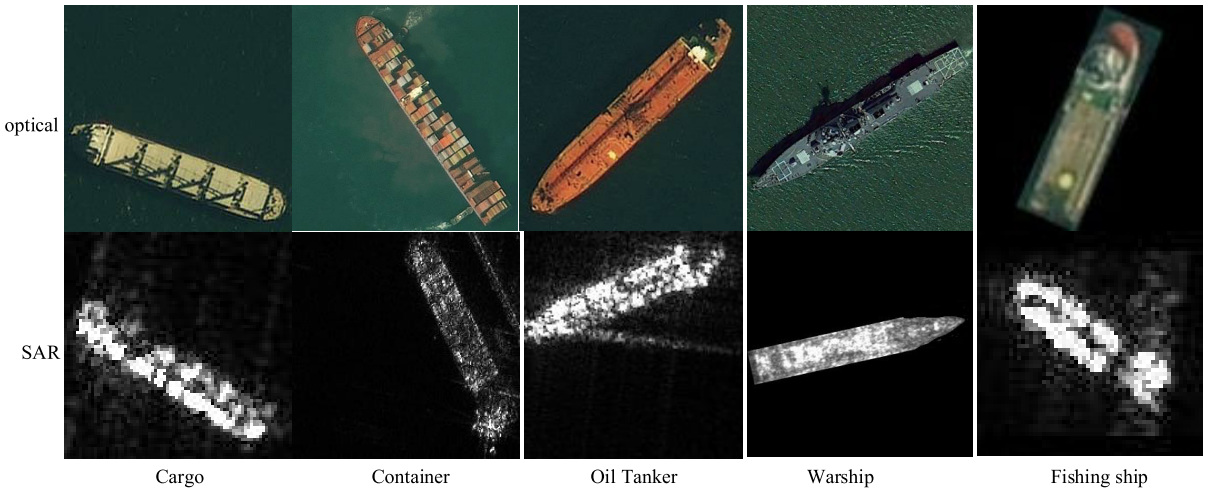


Fig. 5. Examples of the OSship dataset, which includes unpaired optical and SAR ships with five categories.

C. Theoretical Understanding

Following the classical domain adaptation theory, we theoretically analyze our method by analyzing the error bounds of the two stages of the proposed method.

1) First Stage: Motivated by Ben-David et al. [50], the first stage of our method is the analysis of the error bound based on the $\mathcal{H}\Delta\mathcal{H}-$distance of a hypothesis h. The error bound is as follows:

$$\varepsilon_T(h) \leqslant \varepsilon_S(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(S, T) + \lambda. \tag{13}$$

The error bound is decided by the expected error $\varepsilon_T(h)$ of a hypothesis h on the target domain by using three terms: 1) expected error h on the source domain, $\varepsilon_S(h)$; 2) $\mathcal{H}\Delta\mathcal{H}$ − distance, measuring the domain discrepancy of two distributions; and 3) the error λ of the ideal joint hypothesis $h^* = \min_h \varepsilon_S(h) + \varepsilon_T(h)$, which is the optimal classifier on both source and target domains.

Most domain-level domain adaptation methods treat the third term as a negligibly small constant and ignore it. However, due to the huge difference between the optical and SAR domains, features extracted by traditional methods are biased toward the optical domain and incur a high error rate, implying worse discriminability.

To mitigate this pitfall of traditional domain adaptation, the proposed FCDA additionally combines the reconstruction and the consistency constraints of different perturbed versions of the same image to learn the features of unbiased toward specific domain, implicitly reducing the domain characteristics and error rate.

2) Second Stage: The theoretical analysis of the second stage is the extension of Pan et al. [17] and Ben-David et al. [50]. FECA conducts training using a combination of labeled source data and target samples with pseudo-labels. The classification error is calculated as the linear weighted sum of errors in both the source and target domains. Denote y^s and y^t as the labels of the source domain and pseudo-labels of the target domain, respectively. The error can be formalized as follows:

$$\varepsilon_r(h) = \gamma \varepsilon_t(h, \hat{y}^t) + (1-\gamma)\varepsilon_s(h, y^s) \tag{14}$$

where γ is the tradeoff parameter.

The expected error $\varepsilon_t(h, y^t)$ evaluates the classifier learned on the target true labels y^t. The closer the errors of $\varepsilon_t(h, y^t)$ and $\varepsilon_r(h)$ are, the more effective the domain adaptation performs. The following lemma demonstrates that the difference can be bounded for our FECA. The bound is as follows:

$$|\varepsilon_\gamma(h) - \varepsilon_t(h, y^t)| \leqslant (1-\gamma)\left(\frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(S, T) + \lambda\right) + \gamma\rho \tag{15}$$

where ρ is the noise ratio of target pseudo-labels.

In our FECA, the first term $d_{\mathcal{H}\Delta\mathcal{H}}(S, T)$ is evaluated by the class-level discrepancy of instances of the same classed across domains, which relied on the labels and pseudo-labels. Due to the FCDA, following the most relevant cases for domain adaptation in the CV field, the second term can be treated as a negligibly small constant and ignored it. Furthermore, in each iteration, FCDA enhances the representation ability of target domain features and the correctness of pseudo-labels through self-learning, pseudo-label filtering strategy based on prior information and supplementary labels. Therefore, FECA is constantly close to the error bound.

IV. Experiments and Analysis

A. Dataset Description and Setup

In this section, we conduct the UDA from the optical to SAR data for the ship classification task on two datasets. The first dataset is the custom-built "OSship" dataset and the second dataset is public "ORS" dataset. In "OSship," the optical dataset is collected from the subset of FAIR1M [51] and FGSCR [52], while the SAR dataset is collected by the Chinese Gaofen-3 satellite. Some examples of OSship are shown in Fig. 5. Our dataset and code will be available in https://github.com/YuShi1213/Opt2SAR-Unsupervised-Ship-Classification.

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TABLE I
STATISTICS INFORMATION OF FIVE SHIP CLASSES IN OSSHIP

Dataset	Cargo	Container	Oil-Tanker	Warship	Fishing Ship
Optical	372	232	304	304	350
SAR-train	180	118	76	70	110
SAR-test	122	60	63	66	78

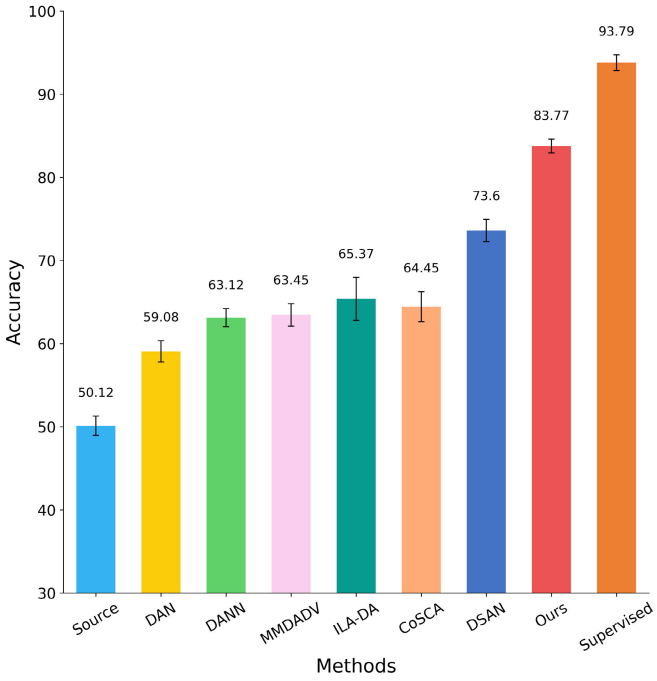


Fig. 6. Accuracy performance of different comparison methods on OSship dataset (AVG ± STD).

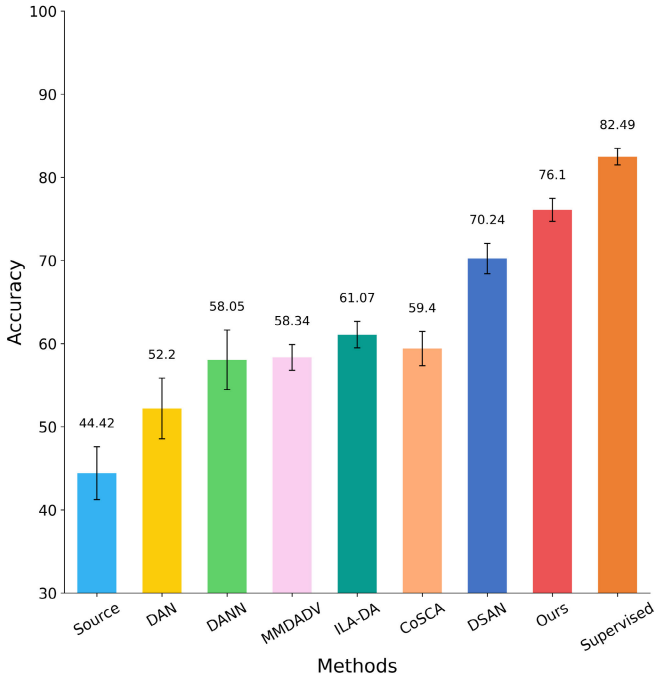


Fig. 7. Accuracy performance of different comparison methods on the ORS dataset (AVG ± STD).

1) Optical Ship Data: Optical ship data are collected from the subset of FAIR1M and FGSCR with the resolution ranging from 0.3 to 0.8 m, which has five ship categories, including "Cargo" with ships transporting different types of cargo, "Container," "Oil-tanker," "Warship" with ships used for military purposes (e.g., cruisers, destroyers, and aircraft carriers), and fishing ship. The number of images per category is shown in Table I.

2) SAR Ship Data: SAR ship data are collected by the Chinese Gaofen-3 satellite with the resolution ranging from 0.5 to 1.8 m, which has the same categories as optical ship data. Part of the SAR ship data comes from FuSAR [54] and has been carefully interpreted by multiple experts to remove low-quality and incorrectly labeled slices. To be consistent with the optical data, the SAR data are carefully processed to ensure that the slice only contains ships and no other clutter. The number of images per category is shown in Table I.

"ORS" contains four ship classes, i.e., cargo, bulk carrier, container, and oil tanker. More details can be found in [29].

The implementation details of the proposed method are described as follows. The learning rate of our method is set to 0.001 with a decay of 0.1. For the convenience of comparison, we utilize ResNet50 [55] as the encoder model since ResNet is the most popular unsupervised domain adaptative classifier in computer vision and SAR target recognition. The batch size of each mini-batch is configured to 48, and the model undergoes 6000 iterations for the first stage and 10 000 iterations for the second stage. The confident threshold of pseudo-labels is set as 0.9. The size upper bound of fishing ship is set as 80 m, while the lower bound of container ship and warship is set as 200 and 150 m, respectively. We project size to image pixel based on image resolution. To allow these samples in the size boundary to participate in model training, we only implement size-based pseudo-label filtering in the first 5000 steps of the second stage. Our method is developed using the Pytorch 1.8 framework on a computer running Ubuntu 20.04, equipped with an Intel Xeon[1] E5-2620v4 CPU and an NVIDIA GTX 3090Ti GPU.

B. Performance Comparison

1) Compared With Other Methods on "OSship": To comprehensively evaluate the effectiveness of our proposed method, we conducted a comparison with typical UDA methods. The comparison methods are detailed in Table II. All comparison methods use ResNet50 as the encoder, and the test network structure of all methods is the same during the

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TABLE II
COMPARISON METHODS

	Methods	Description
Baseline	Source	'Source' is trained only with the labeled optical ship images without any transfer solution, where the encoder is ResNet50. We test the 'Source only' on the test images of SAR domain.
Domain-level	DAN [22]	DAN is a typical distance-based UDA method that aligns the global distributions between two domains. The core idea is that DAN embeds the representations into the reproducing kernel Hilbert space and minimizes the maximum mean discrepancy of two distributions across domains, i.e. MMD distance.
	DANN [25]	DANN is a typical adversarial-based UDA method that aligns the global distributions between two domains. The core idea is that DANN confuses the features to achieve distribution alignment by using the generative adversarial network. DANN proposes the gradient reversal layer (GRL) for end-to-end adversarial training by inserting it between the encoder and discriminator instead of alternating iterative training.
	MMDADV [24]	MMDADV is a UDA practice of SAR target classification that aligns features among the labeled simulated to unlabeled real data. MMDADV uses both the MMD and GRL to achieve global distribution alignment.
Class-level	ILA-DA [19]	ILA-DA achieve class-level alignment by building the constructing contrastive learning with affinity matrices for two domains, where the category information of the target domain is predicted by the source model. To avoid the negative perception caused by wrong pseudo-labels to the network, ILA-DA select the top 75% of target samples in a mini-batch ranked based on their prediction confidence.
	CoSCA [53]	CoSCA performs the class-level alignment by utilizing the triplet loss to achieve class alignment, where the category information of the target domain is predicted by the source model. To improve the quality of pseudo-labels, CoSCA applies two MLP classifiers and combines the result of two MLP classifiers to fuse the predictions for the target domain images.
	DSAN [15]	DSAN maps the feature into class subdomains and achieve the class-level alignment by minimizing the local MMD (LMMD) distance of each class between two domaina, where the category information of target domain is embedded into the LMMD in a weighted manner through the output probabilities of the classifier.
Upperbound	Supervised	'Supervised' supposes that the labels of the target domain can be accessed, which is trained with a fully supervised manner. The encoder is ResNet50.

inference process. "Source only" refers to directly applying the ship classification model trained on the optical domain to classify test images in the SAR domain, which can serve as a baseline. "deep adaptation networks (DAN)" represents the method that aligns the global distribution using MMD constraints, "DANN" signifies the method that aligns the global distribution with an adversarial learning manner, and "MMDADV" represents the methods that align the global domain distributions between the simulated SAR and real SAR data with both the MMD and adversarial constraints. In addition, "deep sub-domain adaptation network (DSAN)," "contrastively smoothed class alignment (CoSCA)," and "instance level affinity domain adaptation (ILA-DA)" refer to class-level alignment approaches. "Supervised" indicates a supervised SAR ship classifier trained with fully labeled SAR training images (the upper bound).

Fig. 6 presents the accuracy performance of various domain adaptive methods on the OSship dataset. The accuracy (ACC) in this article, defined as the ratio of correctly classified ships to the total number of ships, is utilized for assessing the overall recognition accuracy. The "Source" model evaluates the domain gap between the optical and SAR domain and its classification performance serves as a lower bound (baseline) of all the UDA methods. Not using any domain adaptation techniques makes the performance of the "Source" model very poor. All UDA methods show significant improvement compared to baseline (+8.94% +33.65%, respectively), highlighting the benefits of aligning distributions between optical and SAR domains. Domain-level UDA methods, such as "DAN" and "DANN," exhibit modest performance enhancement as they do not account for class information of target domain. "MMDADV" combines techniques from "DAN" and

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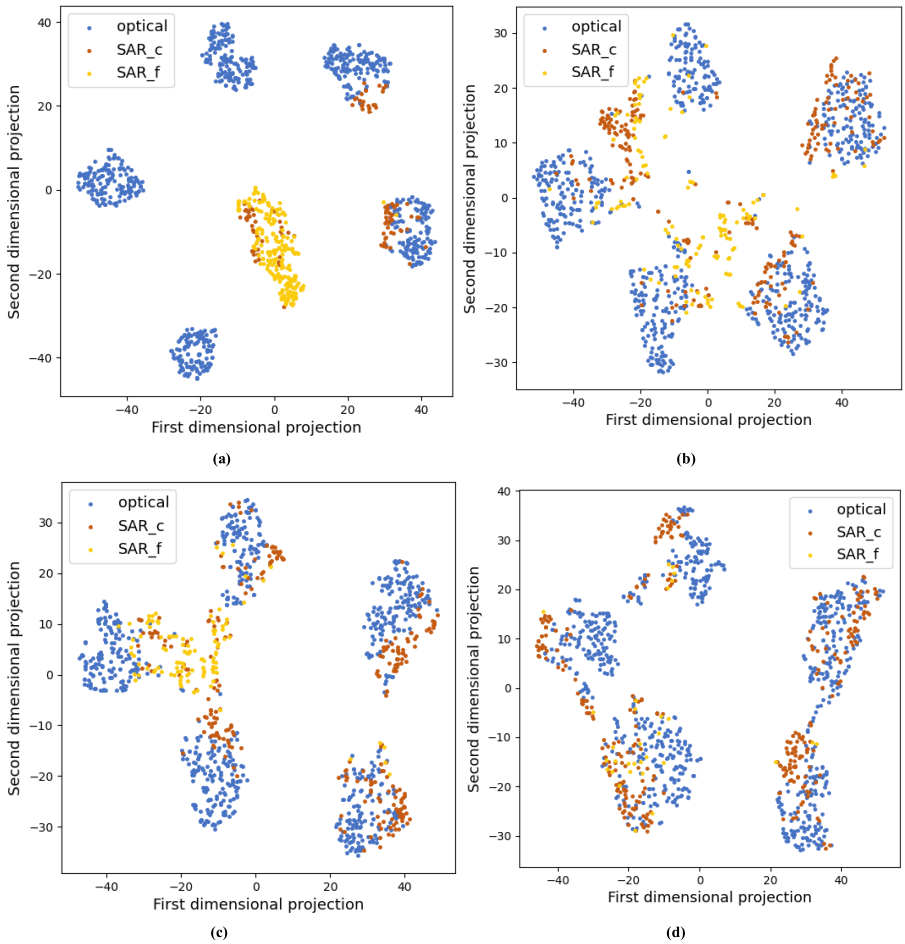


Fig. 8. t-SNE visualization of test split features generated by (a) "Source," (b) "DANN," (c) "DSAN," and (d) proposed approach (ours), where different colors are used to distinguish different situations. Red represents features of correctly aligned SAR images, yellow represents features of falsely aligned SAR images, and blue represents features of optical images.

"DANN" to achieve further performance gains. "ILA-DA" and "CoSCA" outperform others by considering fine class-level information. Notably, instead of using hard pseudo-labels, DSAN incorporates category information from the target domain in a weighted manner through output probabilities of the classifier, leading to superior performance compared to other methods in the comparison. Addressing the significant challenges of optical to SAR UDA, specifically the optical-bias problem stemming from the distinct imaging mechanisms of optical and SAR images as well as the fine-grained class alignment issue, our approach progressively reduces the domain gap at the domain and class levels, which enables effective knowledge transfer for optical-to-SAR classification. Our method demonstrates the highest recognition accuracy among all comparison methods, showcasing its advancement. Despite its excellent performance, our method still lags largely behind the performance fully supervised model (upper bound), which also shows that there is still abundant space to explore in optical-to-SAR UDA.

2) Compared With Other Methods on "ORS": Fig. 7 presents the accuracy performance of various domain adaptive methods on the "ORS" dataset. Since ORS contains less SAR data available, the performance of the compared methods all decreases to varying degrees compared with the OSship dataset. The "Source" model evaluates the domain gap between the optical and SAR domains and its classification performance serves as a lower bound (baseline) of all the UDA methods. Not using any domain adaptation techniques makes the performance of the "Source" model very poor. All UDA methods show significant improvement compared to the baseline, highlighting the benefits of aligning distributions between optical and SAR domains. Our method demonstrates the highest recognition accuracy among all comparison methods, showcasing its advancement.

3) Feature Alignment t-SNE Visualization on "OSship": In order to demonstrate the cross-domain distribution alignment effects of different methods, Fig. 8 displays the t-distribution stochastic neighbor embedding (t-SNE) visualization of features generated by "Source," "DANN," "DSAN," and proposed approach, where different colors are used to distinguish different situations. In Fig. 8(a), it is evident that the SAR features exhibit a wide distribution and show significant domain differences from optical features. Since the optical data are learned in a supervised manner, the interclass discrepancy is very compact and the intraclass discrepancy is very separate, while the SAR features learned by the "Source" exhibit irregular phenomena. By introducing domain adaptation components in the "DANN" and "DSAN" methods, as illustrated in Fig. 8(b) and (c), the domain gap between the optical and SAR distributions is effectively

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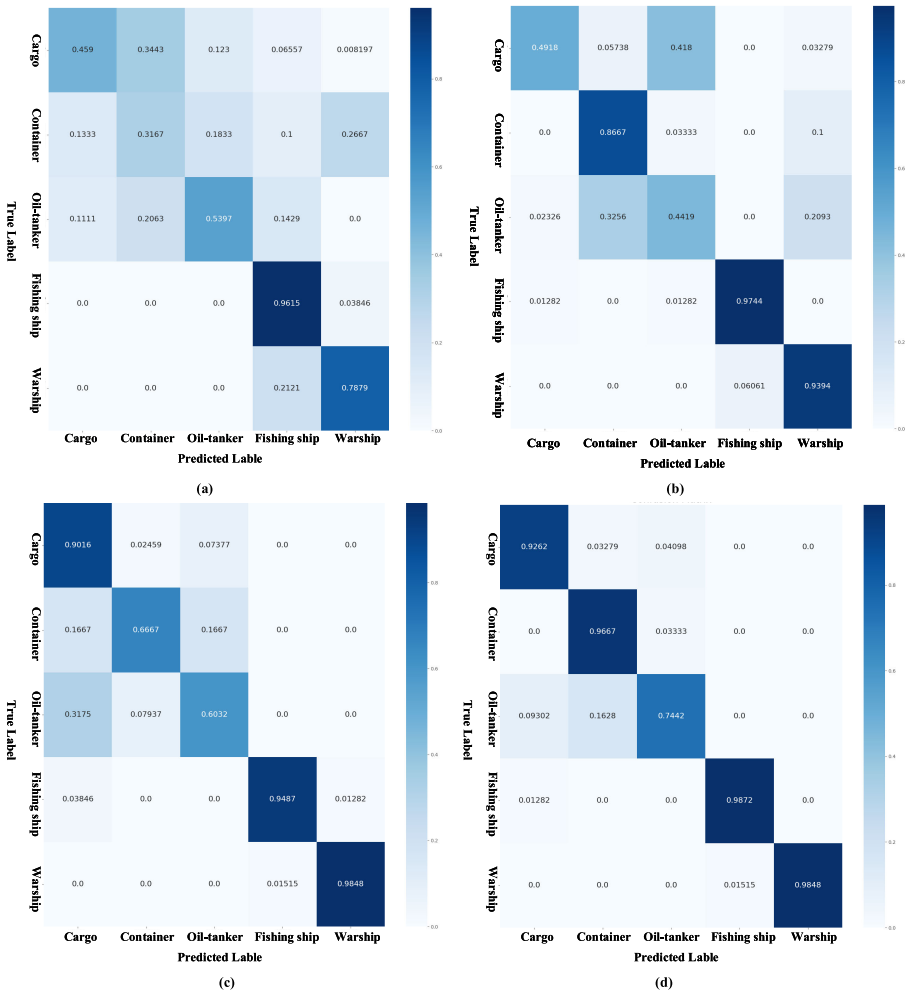


Fig. 9. Confusion matric of the different methods. (a) DANN. (b) DSAN. (c) Ours. (d) Supervised.

reduced. Though the class structure of the optical domain is degraded, the separability of SAR features and their affinity to optical features are improved. Addressing the challenges of optical to SAR UDA in domain level and class level, respectively, specifically the optical-bias problem stemming from the distinct imaging mechanisms of optical and SAR images, as well as the fine-grained class alignment issue, our approach results in more compactly aligned distributions between the optical and SAR features.

4) Confusion Matric: We utilized a confusion matrix analysis method to delve into the reasons why the proposed approach outperforms all other methods under comparison. In Fig. 9, we have illustrated the confusion matrices of select methods as a demonstration. Observing the classification performance of a specific ship class, we find that all methods perform better for classifying the warship and fishing ship, even the simplest DANN. We guess the reason behind this result is that the shape and the aspect ratio of these two categories are obviously different from other categories no matter in the optical or SAR domain. The confusion among cargo, container ship, and oil tanker is more serious since the discriminative features vary in the optical and SAR domains. Among all UDA methods, DANN exhibited the poorest performance, showing significant confusion among the three types of ships. Our method achieves the best performance in terms of comprehensive indicators. Although the classification performance of our method in container ships is not better than DSAN, the classification performance of our method in other categories far exceeds or is equivalent to DSAN. Except for the container ship classification performance, our method closely approached the performance of the supervised method (upper bound). In general, the proposed method learns better feature representation through progressive domain alignment and class alignment, resulting in more compact intraclass features and well-separated interclass features. This facilitates the classifier in establishing improved decision boundaries and effectively separating different ship classes, ultimately enhancing overall classification accuracy.

C. Model Analysis

1) Ablation Study: Built upon the ship classification network, the proposed method consists of the following new modules to solve the opt2SAR UDA ship classification problem, namely, FCDA stage that aims to learn the features of unbiased toward specific domain, implicitly reducing the domain characteristics (i.e., optical-bias characteristics) in the representations, including the DIR module, the perturbed

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TABLE III
ABLATION STUDY (AVG ± STD)

Methods	FCDA			FECA			ACC
	DIR	PR	PC	PosC	CDC	NegC	
Source	-	-	-	-	-	-	50.12 ± 1.16
Ours	✓						63.12 ± 1.10
	✓	✓					67.92 ± 1.08
	✓	✓	✓				69.85 ± 1.28
	✓	✓	✓	✓			79.40 ± 0.96
	✓	✓	✓	✓	✓		81.57 ± 0.92
	✓	✓	✓	✓	✓	✓	$\mathbf{83.77 \pm 0.82}$
Oracle	-	-	-	-	-	-	93.79 ± 0.94

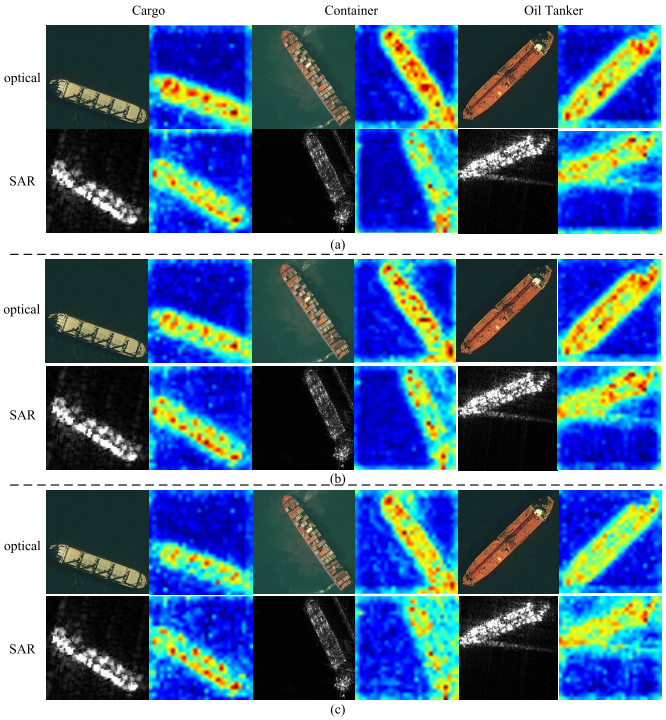


Fig. 10. Typical examples of optical and SAR remote sensing ship images and feature maps for exploring the FCDA. (a) Before domain adaptation. (b) DANN. (c) FCDA.

reconstruction module (PR), the perturbed consistency learning module (PC), FECA stage that aims to learn the finer class information of the SAR domain, including CDC, pseudo-label-based positive classifier consistency learning (PosC) and complementary-label-based negative classifier consistency learning (NegC). To comprehensively evaluate the impact of different modules on ship classification, an ablation study was conducted to assess various versions of the proposed method. The results of the ablation study are depicted in Table III. The results of the ablation study presented in Table III demonstrate that all the new modules contribute to significant performance improvements, confirming their effectiveness. Compared with baseline without transfer learning, only applying DIR constraints achieves limited performance improvement (from 50.12% to 63.12%). The PR encourages the model to focus not only on more discriminative features but also on the information contained in SAR images, achieving performance improvement (from 63.12% to 67.92%). PC encourages the model to be robust to changes in image space so that the model is not limited to focusing only on the regions with more discriminative features, achieving performance further improvement (from 67.92% to 69.85%). Incorporating category information can lead to performance improvements. Combining pseudo-label learning with prior size information notably enhances ship classification performance from 69.85% to 79.40%. The integration of CDC and complementary-labels learning further enhances the class structure, increasing accuracy from 79.40% to 83.77%. In addition, the progressive implementation of FCDA and FECA stages results in significant performance enhancements across all cases, ultimately achieving the best recognition outcomes.

2) Visualization of Domain-Level Alignment: To further examine the effectiveness of FCDA in learning the confused features between the optical and SAR domains, we present the evolutions of the optical and SAR feature maps with the source model, DANN model, and our FCDA model in Fig. 10. In Fig. 9, the first row and the second row represent original optical/SAR ship images and corresponding feature maps of different classes learned by ResNet50 with labeled optical domain. The third row and the fourth row represent the original optical/SAR ship images and corresponding feature maps of different classes learned by DANN with labeled optical domain and unlabeled SAR domain, respectively. The fifth row and the sixth row represent original optical/SAR ship images and corresponding feature maps of different classes learned by FCDA with labeled optical domain and unlabeled SAR domain, respectively. As low-level features have the ability to depict the shape and structural of the object and provide a more intuitive comparison between features, we opted to low-level features for representation. In Fig. 10(a), it is evident that the low-level features without domain adaptation exhibit huge differences between the optical and SAR images. While DANN enforces the model to confuse optical and SAR features with adversarial learning, the distinct imaging mechanisms of optical and SAR images can lead to varying discriminative characteristics of the same classes across

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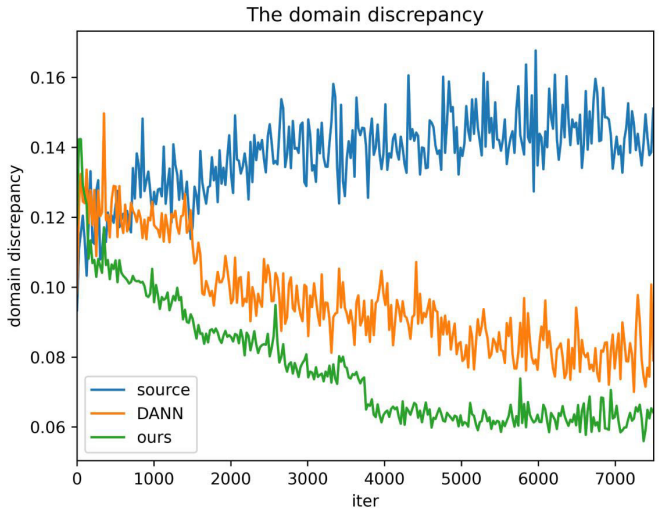


Fig. 11. Domain discrepancy of different methods as the number of iterations changes.

domains. This poses a challenge for DANN in extracting DIR, as the feature representation guided by labeled optical images may be heavily biased toward the source domain. Building upon DANN, FCDA integrates the reconstruction and consistency constraints of different perturbed versions of the same image to learn features that are unbiased toward a specific domain, thereby implicitly reducing domain characteristics (specifically optical-bias characteristics) in the representations. After FCDA, the representations of optical and SAR images exhibit increased confusion, which in turn are propagated to the deeper layers of the model.

To further quantify the reduction of source bias, we introduce the domain distribution discrepancy to measure the reduction. Specifically, the MMD domain distance is used to reflect the source-bias characteristics in the feature representation, that is, the closer the domain distribution distance, the less source-bias features are represented. Fig. 11 gives the domain discrepancy of different methods as the number of iterations changes, where "source" represents the model is trained only with the labeled optical ship images without any transfer solution, "DANN" is a typical adversarial-based UDA method, and "ours" is our proposed FCDA. From Fig. 11, we can see that the domain discrepancy learned by "source" increases with the number of iterations. Since the feature representation guided by the optical label is gradually biased toward the optical domain, the source bias gradually increases. "DANN" and the proposed FCDA both can reduce the optical bias, but compared to "DANN," the proposed FCDA can reduce more effectively.

3) Parameter Analyzation: We further analyze six hyperparameters of our losses for two stages in (7) and (12) on performance. In the process of experiments, we change the value of each hyperparameter while keeping other hyperparameters constant to 1. Tables IV and V present the changes, illustrating the performance of five times. The method's performance shows limited sensitivity to changes in hyperparameter values. In our settings, for simplicity, the values of all hyperparameters are all set as 1. Though they are not the best, they still achieve competitive results.

TABLE IV
ACCURACY UNDER DIFFERENT VALUES OF HYPERPARAMETERS OF STAGE 1

Value	λ_{PR}	λ_{PC}	λ_{DIR}
0.5	70.18 ± 1.02	68.12 ± 1.12	70.42 ± 0.77
1.0	69.85 ± 1.28	69.85 ± 1.28	69.85 ± 1.28
1.5	68.89 ± 1.18	68.85 ± 1.15	70.35 ± 1.33
2	67.82 ± 0.96	68.12 ± 1.11	67.40 ± 0.99
2.5	67.38 ± 1.53	68.38 ± 1.00	68.13 ± 1.85

TABLE V
ACCURACY UNDER DIFFERENT VALUES OF HYPERPARAMETERS OF STAGE 2

Value	λ_{CDC}	λ_{PosC}	λ_{NegC}
0.5	83.13 ± 0.96	82.52 ± 1.08	81.06 ± 1.32
1.0	83.77 ± 0.82	83.77 ± 0.82	83.77 ± 0.82
1.5	84.83 ± 1.02	84.06 ± 0.91	82.78 ± 1.16
2	82.08 ± 0.99	84.83 ± 1.23	82.25 ± 1.18
2.5	81.23 ± 1.26	84.58 ± 1.02	80.86 ± 1.42

D. Discussion

1) Why the Similarity Knowledge Between Domains Should Be Transferred?: The target domain training data under UDA setting are unlabeled, and it is impossible to perform supervised learning directly. Therefore, it needs to transfer the similarity of the two domains to make the labeled optical data guide the learning of unlabeled SAR data. As pointed out by the survey article [13], [14], most UDA methods transfer the commonality/similarity knowledge between the source domain and the target domain to find a good feature representation for the target domain that reduces the error of classification and regression models. Similar to most UDA methods, the transferred knowledge of our methods is the similarity between the optical domain and SAR domain and aims to find a good feature representation for the ship detection task of the SAR domain.

We also give the reason from a probabilistic perspective. The target classification problem can be viewed as learning the posterior probability distribution $P(C|I)$, where I represents the image feature representation extracted by the feature extraction network and C represents the categories. The joint distribution of target classification can be represented by $P(C, I)$, where $P_s(C, I)$ and $P_t(C, I)$ represent the joint distribution of optical data and SAR data, respectively. Due to the difference in imaging mechanisms, the distributions between the source and target domains are different, that is, $P_s(C, I) \neq P_t(C, I)$. According to the Bayesian formula, the joint distribution can be decomposed into the following form:

$$P(C, I) = P(C|I)P(I). \tag{16}$$

Assume that $P(C|I)$ is the same for both domains and the domain shift problem is caused by the marginal distribution $P(I)$. In other words, the ship classifier is consistent between the two domains: given a ship image, no matter which domain the ship image belongs to, the recognition result should be

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the same. To alleviate the domain shift problem, domain adaptation should enforce that the distribution of image representations in the two domains is the same, i.e., $P_s(I) = P_t(I)$. The process of aligning the marginal distributions of two domains is the process of learning the domain invariance between the two domains. However, the cross-domain classifier learned only by using shared features ignores some SAR-specific discriminative features in the process of distribution alignment. The proposed method further combines pseudo-labels and complementary labels to learn the classifier that is more suitable for SAR images.

2) Balance Between Maintaining Domain Invariance and Preserving Target Domain Characteristics: We think that the smaller the difference between domains, the easier it is to maintain the balance between domain invariance and target domain characteristics, and vice versa. We also note that some existing works utilize domain alignment and pseudo-labeling simultaneously across homogeneous domains to maintain a balance between domain invariance and target domain characteristics [33], [53]. However, due to the huge differences in imaging mechanisms between optical and SAR images, achieving the balance between maintaining domain invariance and preserving target domain characteristics directly is difficult, which is verified in some optical-to-SAR cross-domain detection works [13], [36]. The proposed method is to first learn the domain-invariant feature representation and then learn the discriminative feature representation in the target domain. It is a gradual process, that is, first learning a cross-domain shared classifier and then learning an SAR-specific classifier.

3) Resolution Requirement Between SAR and Optical Data: The resolution of optical dataset is 0.3–0.8 m, and the resolution of SAR dataset is 0.5–1.8 m. Our method performs exceptionally well when the resolution disparity between the optical and SAR domains is not substantial. However, if there is a significant resolution difference, such as the SAR domain resolution being 10 m or more, our method may experience a decline in performance. When there is a significant gap in resolution, transferring similarities between two domains becomes challenging and could potentially lead to negative outcomes. If the resolution of the SAR domain is decreased significantly, there will be a noticeable shift in ship size, further increasing the dissimilarity between the two domains. Furthermore, the reduction in SAR image quality will be evident as the ship's shape and other features become distorted.

V. Conclusion

We proposed an innovative UDA method for ship classification that progressively aligns features at domain and class levels by transferring knowledge from labeled optical images to unlabeled SAR images. To address the significant differences between SAR and optical images, we introduced FCDA at the domain level to accurately extract DIR. FCDA, built on an adversarial-learning-based DIR framework, incorporates reconstruction and consistency constraints from various perturbed versions of the same image to prevent bias toward a specific domain, thereby reducing optical-bias characteristics in the representations. At the class level, we introduced FECA to capture finer class information in the SAR domain. FECA includes pseudo-label-based CDC and consistency learning approaches. The CDC models pairwise similarities between domains, where SAR image class information is obtained through pseudo-labels. To ensure the correctness of pseudo-labels, a filtering strategy based on ship size prior information is implemented. We further use the consistency regularization of pseudo-labels and complementary labels to enhance the class structure of unlabeled SAR images. The experimental results indicate that our method achieves exceptional performance in unsupervised classification of SAR ships.

In the future, we will conduct research on open-set domain adaptation. A more practical application scenario is open-set domain adaptation, that is, the source domain and the target domain contain both common target classes and different categories. For open-set domain adaptation, we need to correctly detect or classify the data of known classes in the target domain and reject all data of unknown classes as "unknown" because we have no information about these classes.

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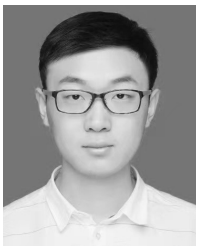
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